Abnormal Uterine Bleeding and Pelvic Pain

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Block: Endocrine/Reproductive/Urologic

Thread: Clinical Medicine

Conflict of Interest: Dr. Cullen has no conflicts of interest to disclose

See page 11 for a helpful list of terms/abbreviations you may not already know.

Further Reading

Practice Bulletin No. 128: Diagnosis of Abnormal Uterine Bleeding in Reproductive-Aged Women. *Obstetrics & Gynecology*. 2012;120(1):197-206. doi:10.1097/AOG.0b013e318262e320 (**Available online to students via the Scott Library**)

Practice Bulletin No. 136: Management of Abnormal Uterine Bleeding Associated With Ovulatory Dysfunction. Obstetrics & Gynecology. 2013;122(1):176-185. doi:10.1097/01.AOG.0000431815.52679.bb (**Available online to students via the Scott Library**)

Learning Objectives

By exam time, you should be able to do the following:

1. Describe normal uterine anatomy
2. Explain normal uterine bleeding patterns as a framework for recognizing abnormal uterine bleeding patterns
3. Discuss causes of abnormal uterine bleeding (AUB) in **reproductive age** people (PALM-COEIN), including clinical presentations, diagnosis and treatment options for each
4. Recognize other (non-uterine) causes of vaginal bleeding
5. Identify common causes of postmenopausal bleeding
6. Describe the imaging modalities commonly used for diagnosing cause of AUB
7. Recognize and describe the common causes of chronic pelvic pain

Study Questions

1. A 38-year-old G1P1 comes to the office complaining of pelvic pain and heavy menstrual bleeding worsening over the last two years. She also reports increasing fatigue and notes that her abdomen feels more distended. You diagnose a pelvic mass on exam. What diagnostic evaluation will you do next?

a. CBC and coagulation studies

b. CBC and pelvic US

c. CT abdomen/pelvis

d. Endometrial biopsy

2. A 14-year-old girl presents with heavy menstrual bleeding. Menarche was one year ago. She reports monthly heavy menses soiling her clothes, and she looks pale. Her mother reports that she bled heavily at the time of tonsillectomy several years ago. Her mother also reports that her own periods have always been heavy, and she hemorrhaged during childbirth. Vitals are normal as is her exam. Labs studies reveal a hemoglobin of 9 and normal thyroid and prolactin tests. What is the most likely cause of her bleeding?

a. Fibroid uterus

b. Immature HPO axis

c. Coagulation disorder

d. Endometriosis

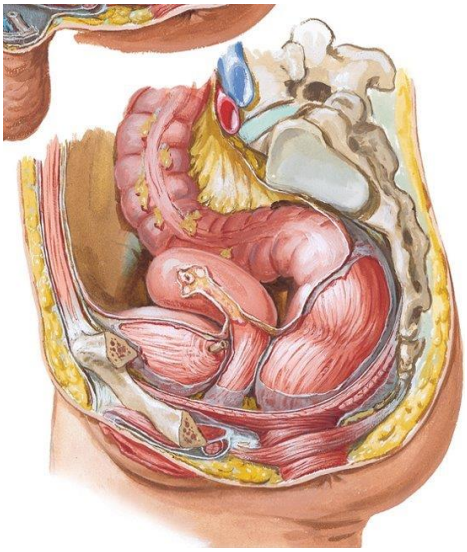
Normal Anatomy of the Uterus

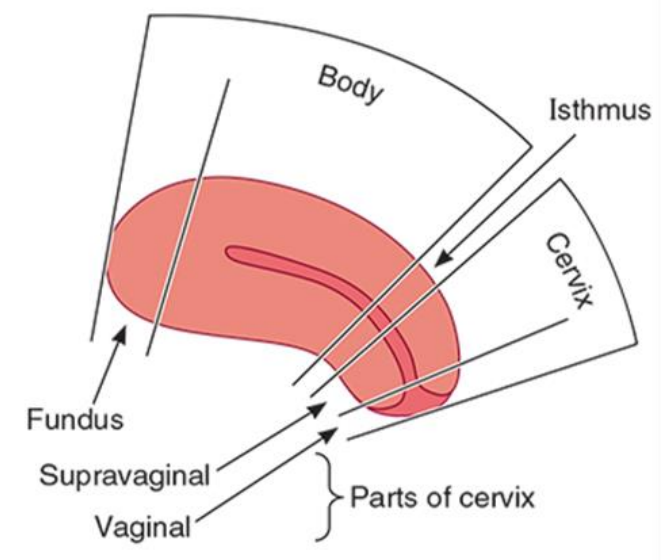
Thick-walled, hollow, muscular organ

Potential space inside the uterus ("endometrial cavity")

Normal nonpregnant uterus lies within the lesser pelvis

- Two parts: body and cervix
- Body lies on the roof of the urinary bladder and the cervix placed between the urinary bladder and rectum

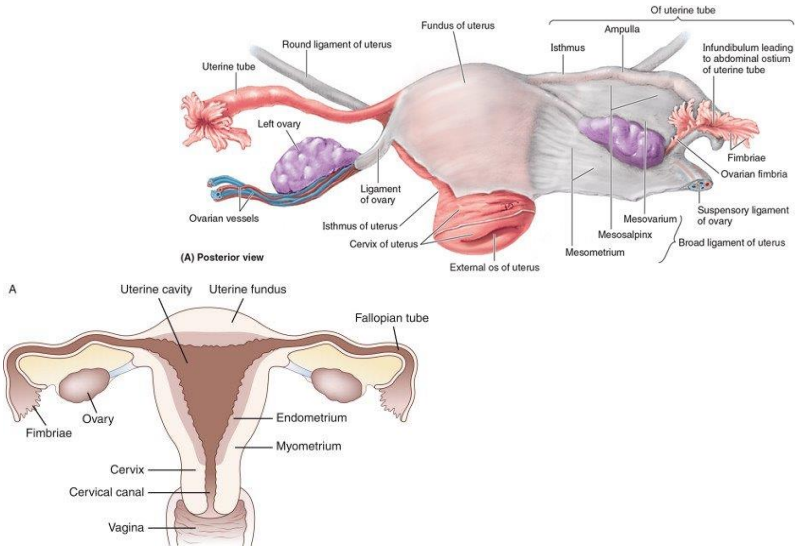




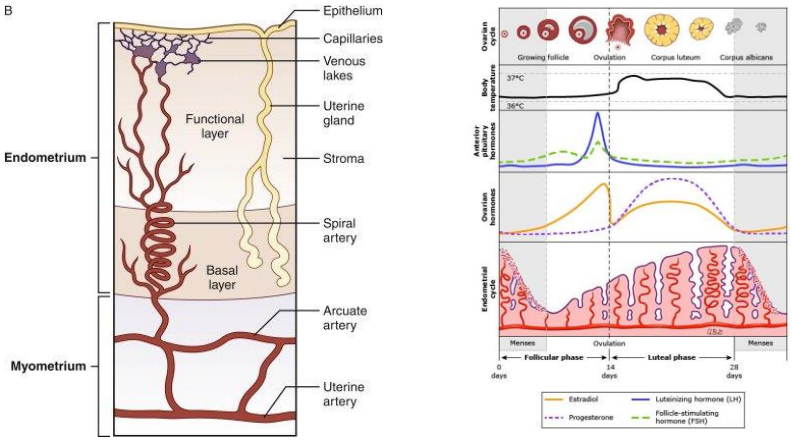
The uterus has three layers "-metrium" refers to the layers of the uterus

- Perimetrium- external - The perimetrium is the serosal layer of the uterus which is derived from peritoneum overlying the uterus.

- Laterally, perimetrium becomes the broad ligament, a double layer of peritoneum. Anteriorly it folds and covers the bladder, forming the vesico-uterine pouch. Posteriorly it reflects and covers the rectum, forming the rectouterine pouch.



- Myometrium- middle layer, smooth muscle layer
  - The myometrium is capable of very strong contractions during labor. Its muscle fibers increase in length and number during pregnancy.
- Endometrium- internal to the myometrium, lines the uterine cavity
  - Divided into basal layer (stratum basalis) and functional layer (stratum functionalis)
    - Functional layer: lining layer of epithelium, venous lakes, capillaries, **uterine glands and stroma**- shed during menses
      - Uterine glands and stroma elsewhere in the body is endometriosis
      - Uterine glands and stroma in the myometrial layer is adenomyosis
    - Basal layer does not shed during menstruation, but its vessels can contribute to normal or abnormal menstrual flow
    - After menses, during the proliferative phase, the basal layer gives rise to a new functional layer. This process is hormonally regulated! You saw this in the "Physiology of the Ovary and Uterus" lecture



Normal vs Abnormal Uterine Bleeding (AUB)

Normal menstrual cycles are between 24-38 days, ≤8 days of bleeding, and ≤7-9 days variation between cycle length, and "normal" flow as determined by the patient.

AUB is anything outside of that in reproductive age people. This includes:

- Amenorrhea (no menses) and infrequent menses (>38-day cycles)
- Frequent menses (<24-day cycles)
- Prolonged menses (bleeding >8 days)
- Irregular cycles (>8 to 10-day variation in cycle length)
- Light menstrual bleeding (uncommon concern)
- Heavy menstrual bleeding **(HMB)**
- Intermenstrual bleeding **(IMB)**

FYI- NOT TESTED- Historical classification you may hear (NO LONGER USED):

a) Menorrhagia = HMB
b) Metrorrhagia = IMB
c) Polymenorrhea = frequent menses
d) Oligomenorrhea = infrequent menses
e) Menometrorrhagia = both heavy and intermenstrual bleeding

Causes of AUB

The mnemonic **PALM –COEIN** lays out the structural vs nonstructural causes of AUB during the **reproductive years** (post-menarche, pre-menopause), with **PALM** being the structural causes and **COIEN** being the non-structural causes.

PALM- Structural causes of AUB

P- Polyp - Hyperplastic overgrowth (typically benign) of endometrial glands and stroma projecting from the surface of the endometrium.

a) Clinical Presentation: May cause HMB, IMB, postcoital bleeding or can be asymptomatic

b) Evaluation: Ultrasound, hysterosalpingogram, sonohysterogram or hysteroscopy. Tissue diagnosis on EMB or D&C.

c) Treatment: Can be removed surgically if symptomatic

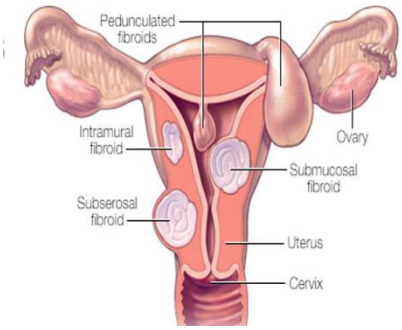
A- Adenomyosis – Endometrial glands and stroma present within the myometrium with resultant hypertrophy of the surrounding myometrium. Etiology uncertain; theories include cell migration after disruption of endometrial-myometrial interface and metaplasia of pluripotent cells

a) Clinical Presentation: Typically, HMB and painful menses (dysmenorrhea). Can also cause dyspareunia and chronic pain.
b) Evaluation: Pelvic exam reveals smoothly enlarged, globular, and sometimes tender uterus. Appearance on pelvic ultrasound or MRI can strengthen your suspicion but definitive diagnosis is made only on surgical pathology examination
c) Treatment: Symptom management with medications (NSAIDs, OCPs, LNG IUDs) or definitive management with hysterectomy (surgical removal of the uterus)

L- Leiomyoma (aka fibroids or myoma) – Benign monoclonal smooth muscle tumors originating from single myometrial cells. Each tumor comes from its own cell, not metastases. Often have somatic mutations and a high concentration of estrogen receptors (hormone responsive). Present in 70-80% of people with a uterus by age 50y. Can occur in different locations within the uterine myometrium including subserosal, intramural, submucosal, and pedunculated (intra-cavitary or subserosal).

a) Clinical Presentation: HMB is caused mainly by fibroids with a submucosal component. Fibroids in other locations may cause pain (dysmenorrhea or dyspareunia) or pressure symptoms, depending on size/location.

b) Evaluation: Pelvic exam reveals enlarged, irregular uterus. Ultrasound or MRI confirm clinical diagnosis.
c) Treatment: Symptom management with OCP, LNG IUD, tranexamic acid, GNRH agonist (depo-Lupron most commonly), OCPs, progesterone. Surgical management with uterine artery embolization (UAE), myomectomy (surgical removal of fibroids) or hysterectomy (surgical removal of uterus).



M- Malignancy or endometrial hyperplasia – endometrial hyperplasia, carcinoma or stromal sarcoma. In premenopausal patient, most often due to unopposed estrogen (as in PCOS with consistent oligoovulation or anovulation)

a) Clinical Presentation: irregular, heavy or prolonged uterine bleeding

b) Evaluation: Endometrial tissue sampling (EMB, D&C, hysteroscopic biopsy)

c) Treatment: hysterectomy, progesterone if poor surgical candidate or fertility desired

COEIN- Non-structural causes of AUB

C- Coagulopathy- Inherited or (rarely) acquired, most commonly VWD

a) Clinical Presentation: Heavy menses since menarche, especially with family history of heavy bleeding

b) Evaluation: labs including coagulation panel, CBC and von Willebrand's testing

c) Treatment: hormonal therapy (OCP, progesterone), tranexamic acid (TXA), Desmopressin (at time of surgery or childbirth)

O- Ovulatory – Ovulatory dysfunction due to medical issue (thyroid disease, premature ovarian failure, hyperprolactinemia, Polyendocrine Metabolic Ovarian Syndrome (PMOS, this was previously known as PCOS), CAH, tumor, extreme calorie restriction, pituitary disease, chemo/radiation) or extremes of age with derangement of hypothalamic-pituitary-axis. If HMB present, mechanism is unopposed estrogen leading to marked thickening of endometrium.

a) Clinical Presentation: Amenorrhea or irregular heavy periods

b) Evaluation: H&P, labs (Beta HCG, prolactin, TSH, FSH, free testosterone, SHBG, estradiol, progesterone, 17-OHP). Consider endometrial sampling.

c) Treatment: OCPs, LNG IUD, cyclic progesterone, weight loss, treat underlying medical issue if present. Insulin sensitizing agents in PCOS. If fertility desired, ovulation induction.

E- Endometrial – usually a diagnosis of exclusion but thought to be due to coagulation defect at the level of the endometrium. No available tests currently.

a) Clinical Presentation: HMB or IMB

b) Evaluation: rule out other causes

c) Treatment: OCPs, TXA, Doxycycline?

I- Iatrogenic – could be due to hormonal medications (such as Depo-Provera, etonogestrel implant (Nexplanon), LNG IUDs), copper IUD or anti-coagulation.

a) Clinical Presentation: HMB or IMB after starting medication

b) Evaluation: medication history, bleeding history

c) Treatment: Wait it out or stop offending medication

N- Not Yet Classified – includes poorly studied entities such as AV malformations, myometrial hypertrophy and any unknown future entities.

Other Causes of Vaginal Bleeding

Non-uterine: vulvar, vaginal, cervical, rectal, urethra/bladder

Postmenopausal bleeding:

a) Atrophy: Hypoestrogenism leads to thinning of the endometrium and vaginal mucosa. Collapsed cells have little fluid to prevent friction damage, resulting in microerosions of the epithelium and inflammation. This tissue then bleeds easily. PE reveals pale, dry epithelium with minimal to no rugation.

b) Cancer/hyperplasia: Thickened endometrium on ultrasound. Diagnosed with tissue sampling.

c) Polyps: benign endometrial or cervical growths.

Imaging

- Ultrasound (abdominal and transvaginal)
  - High frequency sound waves create images
  - may suggest polyp, fibroid, adenomyosis, thickened endometrium in postmenopausal patient (>5mm)
- Sonohysterogram
  - Inject saline through cervix and use ultrasound to visualize intracavitary lesions
- Hysterosalpingogram (HSG)

    - Catheter placed through cervix and contrast introduced into uterine cavity
    - Fluoroscopy (continuous x-ray) used to show the contrast filling the uterus and uterine tubes and spilling out the ends of the tubes
    - Can identify intracavitary structural lesions as well as tubal disease (more on this in your infertility lecture)
- Hysteroscopy
    - Camera passed up the vagina and through the cervix
    - Direct visualization, directed biopsy, and surgical treatment via "operative hysteroscopy" of intracavitary lesions such as polyps and fibroids
- Magnetic Resonance Imaging (MRI)
    - Some role in adenomyosis diagnosis and mapping of fibroids preoperatively

Patient Goals in Treatment

- Improved quality of life/less bleeding
- Contraception vs fertility?
- Medication dosing frequency/ability to comply
- Medical vs surgical treatment
    - Desire to avoid surgery?
    - Desire for definitive treatment?

Chronic Pelvic Pain

6+ months duration of pain in abdomen, pelvic, groin and low back

Causes functional disability

Causes

- Gyn: endometriosis, PID, pelvic congestion, leiomyoma, adenomyosis, malignancy, vulvodynia, vestibulitis, adhesive disease, PID
- Urologic: Interstitial cystitis, bladder pain, recurrent UTI, radiation cystitis, malignancy
- GI: IBS, IBD, diverticular disease, constipation, celiac disease, malignancy
- MSK: pelvic floor dysfunction, DJD, labral tear, arthritis, pelvic girdle pain

- Neuro: nerve entrapment, peripheral neuropathy
- Psych: somatization, physical/sexual abuse, depression

There is overlap between many of these conditions, including IBS, endometriosis, vulvodynia, and myofascial/chronic pain syndromes

Endometriosis: Most commonly found in the ovaries, posterior cul-de-sac, uterosacral ligaments, fallopian tubes, appendix, and round ligament. Can also be found in the lungs, brain, nares, etc. Some interesting clinical scenarios are catamenial hemotysis, penumothorax, nose bleeds or epilepsy (occuring during menstrual bleeds)

Theories on endometriosis include retrograde menstruation, coelemic metaplasia, hematogenous spread, lymphatic spread, stem cell differentiation, and embryogenic theory

Diagnosis is often clinical based on chronic pelvic pain. Physical exam or imaging can help with diagnosis.

Treatment involves NSAIDS, OCPS, progesterone treatment, GnRH antagonists and GnRH agonists, as well as surgical excision.

Glossary

17-OHP- 17-hydroxyprogesterone

CAH- Congenital adrenal hyperplasia

D&C- Dilation and Curettage

Dysmenorrhea- painful menses

EMB- endometrial biopsy

FSH- Follicle stimulating hormone

GC/CT- Gonorrhea and Chlamydia

HCG- Human chorionic gonadotropin

HMB- Heavy Menstrual Bleeding

HPO- Hypothalamic-Pituitary-Ovarian

Hysterectomy- surgical removal of uterus

IMB- Intermenstrual Bleeding

LNG IUD- levonorgestrel intrauterine device

OCPs- Oral contraceptive pills

PMOS- Polendocrine Metabolic Ovarian Syndrome

SHBG- Sex hormone binding globulin

TSH- Thyroid stimulating hormone

TXA- Tranexamic acid

VWD- Von Willebrand Disease

Answers to Study Questions

These answers are not meant to be all encompassing; rather they are examples of the types of thought processes needed for application of the material.

A 38-year-old G1P1 comes to the office complaining of pelvic pain and heavy menstrual bleeding worsening over the last two years. She also reports increasing fatigue and notes that her abdomen feels more distended. You diagnose a pelvic mass on exam. What diagnostic evaluation will you do next?

a. CBC and coagulation studies

b. CBC and pelvic US

c. CT abdomen/pelvis

d. Endometrial biopsy

EXP: this patient most likely has a uterine mass given symptoms and pelvic exam findings. She also has symptoms of anemia, which can become quite severe when it happens over time. She should have a CBC to evaluate for anemia, and pelvic imaging for management discussion. She may ultimately need an endometrial biopsy, but that would not be the first step in this case.

2. A 14-year-old girl presents with heavy menstrual bleeding. Menarche was one year ago. She reports monthly heavy menses soiling her clothes, and she looks pale. Her mother reports that she bled heavily at the time of tonsillectomy several years ago. Her mother also reports that her own periods have always been heavy and she hemorrhaged during childbirth. Vitals are normal as is her exam. Labs studies reveal a hemoglobin of 9 and normal thyroid and prolactin tests. What is the most likely cause of her bleeding?

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EXP: This clinical presentation is consistent with inherited coagulopathy.